

Hw 2 - Warren Montes

3.2) A college of engineering has 7 departments, denoted by a, b, c, d, e, f, and g. Each department has one rep on the student council. From each of these 7 representatives, one is selected to be chair, one to vice-chair, and secretary.
- From Devore 8th ed Ch. 2

How many permutations of size 3 can form?

1. formula

$${}_3P_7 = \frac{7!}{(7-3)!} = \frac{7!}{4!} = 7 \cdot 6 \cdot 5 = 210$$

2. product

7 Choices for chair, once selected, 6 remain for vice-chair, & 5 for secretary
 $\therefore 7 \cdot 6 \cdot 5 = 210$

3. Tree

